

AIM PROTOCOL

Agri • Identity • Microfinance

Version 2.3 — Release Notes & Technical Reference

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Post-Hackathon Development — June 2026

Live: <https://aim-protocol.vercel.app/>

Status	Live on Solana Devnet
Frontend Repo	github.com/amjadkamara/aim-protocol
Smart Contract Repo	github.com/amjadkamara/aim-program
Program ID	AhHHJTu5vodDYE2yLNet2bE6jad9F3xSfbLQdUmykKqB
Hackathon	Solana Frontier 2026 — Submitted V1
V2.3 Branch	v2.3-dev (separate from hackathon submission)

1. Overview

V2.2 delivered the administrative layer — a wallet-gated dashboard giving the platform operator real-time visibility into all registered farmers and loans on Solana devnet. V2.3 builds the multi-stakeholder frontend: the public-facing infrastructure that farmers, lenders, and visitors interact with before any wallet connection is required.

V2.3 is a frontend-only release. The smart contract (Anchor/Rust) is not redeployed in this version. All new pages are built against the existing Program ID and IDL, with contract-dependent features stubbed cleanly and ready to wire to the V2.3 Anchor deployment.

Four major features ship in this version: authenticated home state personalisation, public farmer profile lookup, the loan marketplace, and the full lender registration flow — the most data-rich onboarding form in the protocol to date.

2. Architecture Decisions Locked in V2.3

Before any code was written, the following decisions were debated and locked in. These decisions shape every feature built in V2.3 and beyond.

2.1 Lender Registration Model — Admin-Approved

Farmers register permissionlessly — any wallet can create a FarmerAccount PDA. Lenders require admin approval before they can deploy capital. This gives AIM Protocol a governance layer that NGOs and regulators respect while keeping the farmer experience fully open.

Design Rule:

Farmers: open self-registration. Lenders: admin approval required.

Admin wallet: permanently locked out of both Farmer and Lender roles via require() check at the smart contract level. Role collision is impossible by design.

2.2 Platform Model — Single Infrastructure, Not White-Label

AIM Protocol is a single platform. NGOs and microfinance institutions connect as Lenders — tenants on the protocol. There is no white-label deployment per organisation. This preserves network effects: more lenders compete for farmers on one platform, more farmers build credit histories visible to all lenders.

2.3 Loan Marketplace — Public, No Wallet Required

The marketplace page is publicly accessible. Any visitor — farmer, NGO, journalist, grant committee — can browse active lenders and their terms without connecting a wallet. Wallet connection is only required to apply for a loan. This aligns with the transparency argument of on-chain lending and removes friction for the exact audience AIM Protocol needs to impress.

2.4 Program ID Strategy — Branch Separation

The hackathon submission Program ID (AhHHJTU5vodDYE2yLNet2bE6jad9F3xSfbLQdUmykKqB) is preserved on the main branch and live Vercel deployment. V2.3 frontend development happens on the v2.3-dev branch. When the V2.3 Anchor contract deploys with a new Program ID, the frontend config.ts will be updated in one place — no hardcoded addresses anywhere in the codebase.

3. What V2.3 Built

3.1 Authenticated Home State

File: src/App.jsx

The home screen now has two distinct states — unauthenticated and authenticated — rather than showing marketing content to all users regardless of wallet connection.

State	What the User Sees
Not connected	Full marketing page — hero, How It Works, Features, CTA section, wallet connect button
Connected, farmer PDA found	Welcome back, [Farmer Name] — crop type and district shown beneath. Action buttons only. No marketing copy.
Connected, no farmer PDA	Welcome to AIM Protocol — prompt to create a Farmer ID. Action buttons only.

The farmer name is fetched silently using `fetchFarmer()` on wallet connection. A spinner shows during the fetch. The marketing sections (How It Works, Features, CTA) are wrapped in `{!connected && (...)}` and hidden entirely once a wallet connects. The stats bar (Network, Smart Contract, Built In, Hackathon) remains visible in both states as a trust signal.

- Browse Loan Marketplace button visible to unauthenticated visitors — public page accessible without wallet
- Register as Lender button visible to unauthenticated visitors — lender onboarding starts without wallet, prompts on arrival
- Marketplace and Register as Lender buttons also present in authenticated button group

3.2 Public Farmer Profile Lookup

File: src/FarmerProfile.jsx (NEW)

A public read-only page that allows anyone to look up any farmer's on-chain identity by wallet address. No wallet connection required to view. The page derives the FarmerAccount PDA from the entered address and fetches directly from Solana devnet.

Feature	Detail
Input	Wallet address text field — validates as Solana public key before fetch
Farmer data shown	Name, crop type, district, farm size, member since date, Farmer

	PDA
Loan data shown	Current loan status badge (Active / Overdue / Repaid / No Loan), amount, purpose, due date, requested date
PDA link	Farmer PDA address links directly to Solana Explorer on devnet
Error states	Invalid wallet address, no farmer found — clear plain English messages
Empty state	Clean prompt before any search — 'Enter a wallet address above'
Wallet awareness	Page works fully without connection. Does not block or prompt wallet connection.

Tested live against Binta Marie Winner (wallet: 2TDJWeL4wzGYvSEddbgnTuJJuUbjpreS7zvEPPrkjb1) — profile loads correctly showing Rice, Moyamba, 11 acres, Member Since 2 Jun 2026, PDA address confirmed on Solana Explorer.

Implementation Note:

The existing `fetchFarmer()` and `fetchLoan()` helpers in `useAimProgram.js` only operate on the connected wallet. `FarmerProfile.jsx` fetches directly via `program.account.farmerAccount.fetch(pda)` using an arbitrary public key — no changes to `useAimProgram.js` required.

3.3 Loan Marketplace

File: `src/LoanMarketplace.jsx` (NEW)

A public-facing page listing all approved lenders on the protocol. Built with a read-only Anchor provider that requires no wallet connection — fetches `LenderAccount` PDAs directly from devnet using a dummy wallet pattern.

Feature	Detail
Access	Fully public — no wallet required to browse
Lender cards	Organisation name, wallet address (shortened), Active/Paused badge, max loan, interest rate, duration
Stats strip	Total lenders, active count, paused count — shown when lenders exist
Empty state	Clean 'No lenders registered yet' message with Register as Lender link
CTA — not connected	'Ready to apply for a loan?' — Select Wallet button
CTA — connected	'You're connected' — Go to Dashboard button

Future-ready	Lender cards fully built — populates automatically when V2.3 contract deploys LenderAccount
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Read-Only Provider Pattern:

A dummy wallet object (publicKey: null, signTransaction: passthrough) is passed to AnchorProvider alongside the live RPC connection. This allows program.account.lenderAccount.all() to be called without any wallet — the standard approach for public read-only on-chain data.

3.4 Lender Registration Flow

File: src/LenderRegistration.jsx (NEW)

A three-step wallet-gated registration form for organisations applying to become approved lenders on AIM Protocol. The most data-comprehensive component in the protocol to date — captures everything needed for real-world MFI or NGO onboarding.

Step 1 — Organisation Details

Field	Required	Notes
Organisation Name	Yes	Displayed publicly on the Loan Marketplace
Organisation Type	Yes	NGO / Cooperative / MFI / Development Bank / Impact Fund / CDO / Other
Registration Number	No	Government, business, or NGO registration number — verified by admin
Country of Operation	Yes	15 African countries listed
City / Town	Yes	Registered office location
Website	No	Optional — increases farmer trust on marketplace listing
Contact Email	Yes	Used by admin for approval communication only — not shown publicly
Organisation Description	Yes	Minimum 20 characters — shown to farmers on the marketplace. Live character counter.

Step 2 — Loan Terms & Capital

Field	Required	Notes
Max Loan Amount (SOL)	Yes	Per-farmer ceiling — shown transparently on marketplace card
Interest Rate (% APR)	Yes	Annual rate — cannot be changed post-registration without admin approval

Max Duration (weeks)	Yes	Aligned to farming seasons — max 52 weeks
Min Credit Score	No	Default 0 = accept all farmers. V2.3 credit scoring hooks in here.
Capital Budget (SOL)	Yes	Self-declared total deployment budget. On-chain verified in V2.4 via USDC escrow.
Crop Focus	No	Multi-select tag toggle — Rice, Cassava, Maize, Groundnut, Sweet Potato, Sorghum, Millet, Cocoa, Coffee, All Crops

Step 3 — Review & Submit

- Full data review card — all Step 1 and Step 2 fields displayed before submission
- Expandable AIM Protocol Lender Participation Terms — 10 clauses covering eligibility, admin approval, protocol fee (0.5-1%), capital transparency, loan terms, farmer protection, on-chain records, data reporting, suspension, and governing law
- Two agreement checkboxes — terms acceptance and capital confirmation (both required to enable submit)
- Approval notice — clearly states registration requires admin review before marketplace activation
- Protocol fee disclosed — 0.5-1% per loan disbursement stated explicitly before submission

Additional States

State	Trigger	What is Shown
Wallet Required	Page loaded without connected wallet	Shield icon, explanation that wallet becomes lender identity, Select Wallet button
Already Registered	Connected wallet has existing LenderAccount PDA	Organisation name, active/pending status, wallet address
Pending Approval	Successful form submission	Org name, reg number, location, loan terms, lender wallet, yellow notice to share wallet with admin

Contract Stub:

The handleSubmit function simulates a 1.8s transaction delay and sets status to 'pending'. When the V2.3 Anchor contract deploys with register_lender instruction, replace the stub with:

```
await program.methods.registerLender(...args).accounts({...}).rpc()
```

3.5 App Navigation Updates

File: src/App.jsx (UPDATED)

Change	Detail
New imports	FarmerProfile, LoanMarketplace, LenderRegistration, Store icon, Building2 icon, MapPin icon, Loader2 icon
New routes	farmerprofile, marketplace, lenderregistration
Unauthenticated buttons	Browse Loan Marketplace and Register as Lender visible before wallet connection
Authenticated buttons	Marketplace and Register as Lender added to connected wallet button group
Home state logic	useEffect fetches FarmerAccount PDA on connect — drives personalised greeting

4. Technical Notes

4.1 Read-Only Anchor Provider Pattern

Three pages in V2.3 need to read on-chain data without requiring a wallet connection: FarmerProfile, LoanMarketplace, and the existing-registration check in LenderRegistration. The pattern used is a dummy wallet object passed to AnchorProvider:

```
const dummyWallet = {
  publicKey: null,
  signTransaction: async (tx) => tx,
  signAllTransactions: async (txs) => txs,
}
const provider = new AnchorProvider(connection, dummyWallet, { commitment: 'confirmed' })
const program = new Program(idl, provider)
```

This creates a valid Anchor program instance capable of all read operations without signing capability. Fetch calls succeed; any instruction that requires signing will correctly fail — which is the intended behaviour for public read-only pages.

4.2 Icon Input Padding — Inline Style Override

Tailwind's JIT compiler does not always generate padding utility classes reliably when composed dynamically. All icon inputs in LenderRegistration.jsx use a CSS custom property via inline style rather than a Tailwind class:

```
const ICON_PADDING = { paddingLeft: '2.5rem' }

// Applied directly to each icon input:
<input className={baseInput} style={ICON_PADDING} />
```

Inline style always wins over Tailwind in the cascade — this guarantees the icon never overlaps the placeholder text regardless of build configuration.

4.3 getLoanStatus() in FarmerProfile

FarmerProfile.jsx uses the same defensive getLoanStatus() helper pattern established in AdminDashboard.jsx — Object.keys(loan.status)[0] wrapped in try/catch to handle null, undefined, or legacy account shapes. This ensures the profile renders cleanly for all devnet accounts including those created before V2.1.

4.4 Test Data — Fresh Deploy Plan

All existing devnet farmer and loan accounts were created before V2.3 features existed and represent noise data. The V2.3 contract deployment (separate branch) will start with a clean devnet state using the following wallet role separation:

Wallet	Role	Notes
Main wallet (Amjad)	Admin only	Never registers as farmer or lender — enforced at contract level
Binta's wallet	Farmer	Re-registers fresh after V2.3 contract deploy
3rd wallet	Farmer	Second independent farmer for credit score and loan counter testing
4th wallet	Lender	Registers as lender org — tests per-lender dashboard filter

5. File Changelog

File	Status	What Changed
src/App.jsx	Updated	New imports, 3 new routes, authenticated home state with farmer greeting, marketplace and lender registration buttons for both auth states

src/FarmerProfile.jsx	New	Public farmer profile lookup by wallet address — read-only, no wallet required, fetches FarmerAccount and LoanAccount PDAs
src/LoanMarketplace.jsx	New	Public lender marketplace — read-only Anchor provider, lender cards, wallet-aware CTA, empty state with register link
src/LenderRegistration.jsx	New	3-step lender onboarding form — organisation details, loan terms and capital, review and submit with expandable terms and dual agreement checkboxes
docs/screenshots/17-farmer-profile-lookup/	New	3 screenshots — empty state, Binta profile loaded, invalid wallet error
docs/screenshots/18-authenticated-home/	New	1 screenshot — welcome back greeting with farmer name, crop, district
docs/screenshots/19-loan-marketplace/	New	2 screenshots — no wallet state, wallet connected state
docs/screenshots/20-lender-registration/	New	7 screenshots — wallet required, Step 1 empty and filled, Step 2 empty and filled, Step 3 review, Step 3 terms expanded

6. V2.3 Verification Checklist

Authenticated Home State

- ✓ Wallet connected — marketing sections hidden, welcome state shown
- ✓ Farmer PDA found — Welcome back, [Farmer Name] displayed with crop and district
- ✓ No farmer PDA — Welcome to AIM Protocol prompt shown
- ✓ Marketplace button visible before and after wallet connection
- ✓ Register as Lender button visible before and after wallet connection

Farmer Profile Lookup

- ✓ Binta Marie Winner lookup — profile loads correctly from devnet
- ✓ Invalid wallet address — plain English error message shown
- ✓ Unknown valid address — No farmer profile found message shown
- ✓ Farmer PDA link opens correct account on Solana Explorer
- ✓ Page functions without wallet connection

Loan Marketplace

- ✓ Page loads without wallet connection
- ✓ Empty state shown — No lenders registered yet

- ✓ Unauthenticated CTA — Select Wallet button shown
- ✓ Authenticated CTA — Go to Dashboard button shown
- ✓ Deployed and accessible at aim-protocol.vercel.app

Lender Registration

- ✓ No wallet — Wallet Required screen shown
- ✓ Step 1 validation — Continue disabled until all required fields complete
- ✓ Description character counter — red below 20, green above
- ✓ Step 2 — all icon inputs clear of placeholder text (inline style fix)
- ✓ Crop focus toggle — multi-select works, green highlight on selected
- ✓ Step 3 review — all Step 1 and Step 2 data displayed correctly
- ✓ Terms expand/collapse — scrollable 10-clause terms visible
- ✓ Submit disabled until both checkboxes checked
- ✓ Submit stub — simulates tx delay, shows Pending Approval screen
- ✓ Pending screen — org name, reg number, location, loan terms, wallet address shown
- ✓ Committed to GitHub and deployed on Vercel

7. V2.3 Smart Contract Roadmap

The following Anchor/Rust changes are required to complete V2.3. All frontend components are built and waiting — the contract is the only remaining blocker.

Instruction / Change	Purpose	Frontend Impact
register_lender	Creates LenderAccount PDA per lender wallet with org name, terms, status flags	LenderRegistration.jsx — replace stub with real call
approve_lender (admin-only)	Flips lender isActive flag — only admin wallet can call	Admin dashboard lender approval UI (V2.3 admin update)
LenderAccount struct	Stores: name, orgType, country, city, email, maxLoanSol, interestRateBps, maxDurationWeeks, minCreditScore, capitalBudget, cropFocus, isActive	LoanMarketplace.jsx — populates automatically
FarmerAccount update	Add loan_counter: u64 and credit_score: u64 fields	FarmerProfile.jsx credit score display — field already in component, awaiting data
LoanAccount update	Add lender: Pubkey field to tag each loan with issuing lender	Per-lender dashboard filter — reads lender pubkey from loan
Admin role check	require!(signer !=	Prevents admin wallet from taking

	ADMIN_PUBKEY) in register_farmer and register_lender	farmer or lender roles
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8. V2.4 Roadmap

- USDC stablecoin disbursement — loans in USDC instead of SOL, removing price volatility for farmers
- USDC escrow accounts per lender — capital locked on-chain, verified against declared budget
- Pyth oracle integration — crop price feeds for dynamic loan sizing based on current market value
- Per-lender dashboard — lenders see only their own farmers and loans, filtered by lender pubkey
- Capital budget verification — on-chain check that lender wallet holds declared USDC before approval
- Multi-sig loan approval — cooperative governance for lenders operating as a board
- KYC integration — government ID tied to wallet address for donor and NGO compliance

AIM Protocol V2.3 — Multi-stakeholder frontend complete. Smart contract deployment next.

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